03.

Surds and Indices

Type - 8

1. Solve :-

$$\frac{\sqrt{4375}}{\sqrt{7}} = ?$$

(a) 64 (b) 25
(c) 36 (d) 16

RRB RPF Constable -22/01/2019 (Shift-I)

Ans. (b) Given,

$$\frac{\sqrt{4375}}{\sqrt{7}} = \sqrt{\frac{4375}{7}}$$

$$= \sqrt{625} = \sqrt{25 \times 25} = 25$$

2. The square root of 10201 is:

(a) 91 (b) 99
(c) 101 (d) 111

RRB RPF Constable -18/01/2019 (Shift-I)

Ans : (c) Finding the square root by division method,

	101
1	10201
+1	1
201	×0201
1	0201
	××××

Hence, the square root of 10201 is 101.

3. The square root of 519841 is:

(a) 721 (b) 629
(c) 631 (d) 731

RRB RPF SI -10/01/2019 (Shift-II)

Ans. (a) : Square root of 519841

	721
7	519841
+7	49
142	298
+2	284
1441	1441
1	1441
	××××

Hence, the square root of 519841 is 721.

4. What is the square root of 34596 ?

(a) 174 (b) 176
(c) 204 (d) 186

RRB RPF Constable -19/01/2019 (Shift-II)

Ans. (d) Square root of 34596

	186
1	34596
+1	1
28	245
+8	224
366	2196
6	2196
	××××

So, the required square root of the given number is 186.

5. What is the square root of 11881?

(a) 109 (b) 119
(c) 111 (d) 101

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (a) : Square root of 11881

	109
1	1 1 8 8 1
1	1
209	1 8 8 1
	1 8 8 1
	××××

So, the required square root is 109.

6. Which of the following is the square root of 14161 ?

(a) 129 (b) 121
(c) 119 (d) 131

RRB Group-D – 03/10/2018 (Shift-I)

Ans : (c) Finding the square root of 14161

	119
1	14161
+1	1
21	×41
+1	21
229	2061
	2061
	××××

Hence, 119 is the square root of 14161.

7. What is the square root of 7569 ?

(a) 77 (b) 87
(c) 93 (d) 83

RRB Group-D–03/10/2018 (Shift-II)

Ans : (b) Square root of 7569

```
        87
   8 | 7569
   8 | 64
 167 | 1169
   7 | 1169
     | ××××
```

Hence, the square root of 7569 is 87

8. What is the square root of 8281 ?

(a) 81 (b) 91
(c) 89 (d) 99

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (b) Square root of 8281

```
       91
   9 | 8 2 8 1
   9 | 8 1
 1 8 1 |   1 8 1
         1 8 1
         ××
```

Hence, the required square root is 91.

9. What is the square root of 3364 ?

(a) 64 (b) 62
(c) 58 (d) 52

RRB Group-D – 15/10/2018 (Shift-I)

Ans : (c) Square root of 3364

```
          | 5  8
      5   | 3  3  6  4
+     5   | 2  5
10    8   |    8  6  4
          |    8  6  4
          |    ×  ×  ×
```

Hence, the required square root is 58.

10. What is the value of $\sqrt{183184}$?

(a) 414 (b) 432
(c) 428 (d) 416

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (c) $\sqrt{183184}$ = The square root of 183184.
So finding the square root of 183184,

```
            | 4  2  8
        4   | 1  8  3  1  8  4
    +   4   | 1  6
    8   2   |    2  3  1
+       2   |    1  6  4
8   4   8   |       6  7  8  4
        8   |       6  7  8  4
            |       ×  ×  ×  ×
```

Hence, the required value is 428.

11. The square root of 60025 is:

(a) 275 (b) 255
(c) 245 (d) 265

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (c) : Finding the square root of 60025 by factorization method,

```
5 | 60025
5 | 12005
7 | 2401
7 | 343
7 | 49
7 | 7
  | 1
```

$\sqrt{60025} = \sqrt{5\times5\times7\times7\times7\times7}$
$= 5 \times 7 \times 7$
$= 5 \times 49 = 245$

Hence, the required square root is 245.

12. The square root of 4624 is:

(a) 62 (b) 72
(c) 78 (d) 68

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (d) : Finding the square root by factorization method,

```
 2 | 4624
 2 | 2312
 2 | 1156
 2 | 578
17 | 289
17 | 17
   | 1
```

$\sqrt{4624} = \sqrt{2\times2\times2\times2\times17\times17}$
$= 2 \times 2 \times 17$
$= 4 \times 17 = 68$

Hence, the required square root is 68.

13. What is the square root of 5476?

(a) 84 (b) 74
(c) 66 (d) 76

RRB Group-D – 24/09/2018 (Shift-I)

Ans : (b) Finding the square root of 5476 by division method,

```
       74
   7 | 5476
  +7 | 49
 144 |  576
   4 |  576
     |  ×××
```

Hence, the required square root is 74,

14. The square root of 4356 is:

(a) 76 (b) 64
(c) 84 (d) 66

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (d) : Finding the square root by division method,

```
       66
   6 | 4356
   6 | 36
 126 | x 756
   6 |   756
     |   xxx
```

Hence, the required square root is 66.

15. Find the square root of 15625.

(a) 145 (b) 125
(c) 135 (d) 150

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (b) Finding the square root by division method,

	125
1	$\overline{1}\ \overline{56}\ \overline{25}$
+1	1
22	56
+2	44
245	1225
	1225
	xxxx

$\sqrt{15625} = 125$

Hence, the square root of 15625 is 125.

16. The square root of 10816 is:

(a) 106 (b) 96
(c) 114 (d) 104

RRB Paramedical Exam – 20/07/2018 (Shift-I)

Ans. (d) : The square root of 10816

	104
1	10816
+1	1
20	08
+0	00
204	816
4	816
	×××

Hence, the required square root is 104.

17. Find the value of $\sqrt{37249}$ –

(a) 183 (b) 187
(c) 197 (d) 193

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (d) $\sqrt{37249}$ = square root of 37249.

So, finding the square root,

	193
1	$\overline{37}\overline{249}$
+1	1
29	272
+9	261
383	1149
3	1149
	××××

Hence, the value of given number is 193.

18. If $\frac{x}{\sqrt{243}} = \frac{\sqrt{2187}}{x}$, and x is positive, then what is the value of x ?

(a) 29 (b) 27
(c) 23 (d) 21

RRB Group-D – 12/11/2018 (Shift-II)

Ans : (b) Given,

$$\frac{x}{\sqrt{243}} = \frac{\sqrt{2187}}{x}$$

$$\Rightarrow \frac{x}{\sqrt{9\times9\times3}} = \frac{\sqrt{3\times9\times9\times9}}{x}$$

$$\Rightarrow \frac{x}{9\sqrt{3}} = \frac{27\sqrt{3}}{x}$$

$$\Rightarrow x^2 = 27\times9\times\sqrt{3}\times\sqrt{3}$$

$$x = \sqrt{27\times27} = 27$$

Hence, the value of x is 27.

19. Solve the following equation.

$\sqrt{54}\times\sqrt{6}=?$

(a) 18 (b) 19
(c) 20 (d) 16

RRB Group-D – 23/10/2018 (Shift-III)

Ans : (a) From given equation,

$\sqrt{54}\times\sqrt{6}$

$\sqrt[a]{m}\times\sqrt[a]{n} = \sqrt[a]{m\times n}$ से

$= \sqrt{54\times6}$

$= \sqrt{324}$

$= 18$

20. Which from the following is irrational?

(a) $\sqrt[6]{4096}$ (b) $\sqrt[4]{4096}$
(c) $\sqrt[8]{4096}$ (d) $\sqrt[3]{4096}$

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (c) From options,

$\sqrt[6]{4096} = 4$

$\sqrt[4]{4096} = 8$

$\sqrt[3]{4096} = 16$

$\sqrt[8]{4096} = 2\sqrt[8]{(2)^4}$

Hence, $\sqrt[8]{4096}$ is irrational, because all others are rational.

21. What will be the square root of 25281?

(a) 149 (b) 143
(c) 139 (d) 159

RRB Group 'D' 07/12/2018 (Shift-I)

Ans : (d) finding the square root by division method,

	159
1	$\overline{25}\overline{281}$
+1	1
25	152
+5	125
309	2781
9	2781
	××××

Hence, the required square root is 159.

22. What is the square root of 39204?

(a) 198 (b) 196
(c) 194 (d) 202

RRB Group-D – 07/12/2018 (Shift-III)

Ans : (a) The square root of 39204

	198
1	39204
+1	1
29	292
+9	261
388	3104
8	3104
	××××

Hence the square root of 39204 will be 198.

23. Which of the following is the square root of 35721?

(a) 179 (b) 189
(c) 171 (d) 201

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (b) (a) $179 \rightarrow (179)^2 = 32041$
(b) $189 \rightarrow (189)^2 = \boxed{35721}$
(c) $171 \rightarrow (171)^2 = 29241$
(d) $201 \rightarrow (201)^2 = 40401$

24. The square root of 41616 is:

(a) 196 (b) 204
(c) 186 (d) 194

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (b) Square root of 41616

	204
2	41616
2	4
404	1616
4	1616
	××××

Hence, the square root of 41616 is 204.

25. What is the square root of 16641?

(a) 139 (b) 121
(c) 129 (d) 131

RRB Group-D – 03/12/2018 (Shift-II)

Ans : (c) The square root of $16641 = \sqrt{16641}$
$= \sqrt{3\times3\times43\times43} = 3\times43 = 129$

26. If $\sqrt{108}+\sqrt{243} = 25.98$ find the value of $\sqrt{147}+\sqrt{192}$.

(a) 26.89 (b) 27.172
(c) 25.98 (d) 24.248

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (c) Given,

$\sqrt{108}+\sqrt{243} = 25.98$

$\Rightarrow \sqrt{2\times2\times3\times3\times3} + \sqrt{3\times3\times3\times3\times3} = 25.98$

$\Rightarrow 6\sqrt{3} + 9\sqrt{3} = 25.98$

$\Rightarrow 15\sqrt{3} = 25.98$ .....(I)

According to the question,

$\therefore \sqrt{147}+\sqrt{192} = \sqrt{7\times7\times3} + \sqrt{2\times2\times2\times2\times2\times2\times3}$

$= 7\sqrt{3}+8\sqrt{3} = 15\sqrt{3}$ (From equation- I)

$= 15 \times 1.732$

$= 25.980$

$= 25.98$

27. What is the square root of 243049?

(a) 497 (b) 503
(c) 493 (d) 487

RRB Group-D – 15/11/2018 (Shift-III)

Ans (c) Finding the square root by division method,

$\sqrt{243049} = ?$

	493
4	243049
+4	16
89	830
+9	801
983	2949
+3	2949
	××××

Hence, the required square root is 493.

28. How much is the square root of 21904?

(a) 144 (b) 146
(c) 152 (d) 148

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (d) Square root of 21904

	148
1	21904
+1	1
24	119
+4	96
288	2304
8	2304
	××××

Hence, the required square root is 148.

29. What is the square root of 16129?

(a) 143 (b) 137
(c) 127 (d) 117

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (c) Square root of 16129

	127
1	16129
+1	1
22	61
+2	44
247	1729
7	1729
	××××

Hence, the square root of 16129 is 127.

30. How much is the square root of 10404?

(a) 102 (b) 106
(c) 98 (d) 104

RRB Group-D – 12/11/2018 (Shift-II)

Ans : (a) The square root of 10404,

$\sqrt{10404} = \sqrt{2\times2\times3\times3\times17\times17}$

$= 2 \times 3 \times 17 = 102$

Hence, the square root is 102.

31. How much is the square root of 1521?

(a) 41 (b) 39
(c) 31 (d) 49

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (b) The square root of 1521

	39
3	15 21
+3	9
69	621
+9	621
	×××

Hence, the required square root is 39.

32. The square root of 16900 is:

(a) 130 (b) 110
(c) 140 (d) 120

RRB Group-D – 05/10/2018 (Shift-III)

Ans. (a) Square root of 16900

	130
1	16900
+1	1
23	×69
+3	69
260	××00
+ 0	00
	××

Hence, the required square root is 130.

Type - 2

33. The square root of 13.69 is:

(a) 3.7 (b) 37
(c) 0.037 (d) 0.37

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (a) The square root of 13.69 by division method,

	3.7
3	13.69
3	9
67	469
7	469
	×××

Hence, the required square root is 3.7.

34. $\sqrt{0.00069169} = ?$

(a) 0.00243 (b) 0.000243
(c) 0.0263 (d) 0.243

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (c) : From given number,

$\sqrt{0.00069169} = \sqrt{\frac{69169}{100000000}}$

$= \sqrt{\frac{69169}{100000000}} = \sqrt{\frac{263\times263}{10000\times10000}} = \frac{263}{10000} = 0.0263$

35. Find the square root of 0.0324.

(a) 0.18 (b) 1.8
(c) 1.08 (d) 0.018

RRB Group-D – 24/09/2018 (Shift-I)

Ans : (a) The square root of given number,

$\sqrt{0.0324} = \sqrt{\frac{324}{10000}}$

$= \sqrt{\frac{18\times18}{100\times100}}$

$= \frac{18}{100} = 0.18$

Hence, the required square root is 0.18.

36. Find the value of $\sqrt{4.2436}$

(a) 2.14 (b) 2.16
(c) 2.04 (d) 2.06

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (d) : $\sqrt{4.2436}$ = Square root of 4.2436

	206
2	42436
+2	4
406	×2436
6	2436
	××××

$\sqrt{4.2436} = \sqrt{\frac{42436}{10000}} = \frac{206}{100} = 2.06$

37. Find the value of $\sqrt{0.5}$ –

(a) 0.707 (b) 0.947
(c) 0.787 (d) 0.897

RRB Group-D – 04/10/2018 (Shift-II)

Ans : (a) $\sqrt{0.5}$ = Square root of 0.50000

	0.707
7	0.500000
+7	49
1407	10000
7	9849
	151

Hence, the square root of 0.5 is 0.707 (Approx). So, ($\sqrt{0.5}$ = 0.707)

38. Find the value of $\sqrt{0.6}$ –

(a) 0.944 (b) 0.874
(c) 0.894 (d) 0.774

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (d) : $\sqrt{0.6}$ = Square root of 0.600000
So, finding the square root by division method,

	0.774
7	0.600000
+7	49
147	1100
+7	1029
1544	7100
4	6176
	924

Hence the required value is 0.774 (approx).

39. What is the square root of 15625?
(a) 105 (b) 115
(c) 125 (d) 135
RRB Group-D – 12/11/2018 (Shift-II)

Ans : (c) The square root of 15625 = $\sqrt{15625}$
$= \sqrt{5\times5\times5\times5\times5\times5}$
$= 5\times5\times5 = 125$

40. What is the value of $\sqrt{0.8}$
(a) 0.964 (b) 0.694
(c) 0.894 (d) 0.984
RRB Group-D – 12/10/2018 (Shift-II)

Ans : (c) $\sqrt{0.8}$ = square root of 0.80000
So, finding the square root

	0.	8	9	4
8	0.	80	00	00
+8		64		
169		16	00	
+ 9		15	21	
1784			79	00
4			71	36
			7	64

Hence, the value of $\sqrt{0.8}$ is 0.894.

41. The value of $\sqrt{0.0144}$ is:
(a) 0.12 (b) 0.012
(c) 1.2 (d) 0.0012
RRB NTPC 27.04.2016 Shift : 1

Ans : (a) $\sqrt{0.0144}$ = Square root of 0.0144
So, finding the square root of 0.0144 by division method,

	0.12
1	0.0144
1	01
22	44
+2	44
	××

Hence, the square root of 0.0144 = $\sqrt{0.0144}$ = 0.12

Type - 8

42. If $\sqrt{4225} = 65$. Find the value of $\sqrt{42.25}+\sqrt{0.4225}$.
(a) 6.5 (b) 7.25
(c) 6.25 (d) 7.15
RRB JE - 30/05/2019 (Shift-III)

Ans : (d) Given $\sqrt{4225} = 65$
Then, $= \sqrt{42.25}+\sqrt{0.4225}$
= 6.5 + .65 ..(Square root of 4225 = 65)
= 7.15
Hence, the required value is 7.15

43. If $\sqrt{7}$ = 2.6457 and $\sqrt{3}$ =1.732, Then find the value of $\frac{1}{\sqrt{7}-\sqrt{3}}$
(a) 1.0944 (b) 1.944
(c) 1.009 (d) 1.0844
RRB RPF Constable -25/01/2019 (Shift-III)

Ans : (a) Given,
$\sqrt{7} = 2.6457$
$\sqrt{3} = 1.732$
then,
$$\frac{1}{\sqrt{7}-\sqrt{3}} = \frac{\left(\sqrt{7}+\sqrt{3}\right)}{\left(\sqrt{7}-\sqrt{3}\right)\times\left(\sqrt{7}+\sqrt{3}\right)} = \frac{\sqrt{7}+\sqrt{3}}{4}$$
$$= \frac{2.6457+1.732}{4} = \frac{4.3777}{4} = 1.0944$$
Hence, the required value is 1.0944.

44. If $\frac{0.27}{p^2} = 27$, p = ?
(a) 0.001 (b) 0.1
(c) 0.01 (d) 1.0
RRB RPF SI -13/01/2019 (Shift-II)

Ans. (b) Given,
$$\frac{0.27}{p^2} = 27$$
$$p^2 = \frac{0.27}{27} = \frac{27}{2700}$$
$$p^2 = \frac{1}{100} = \left(\frac{1}{10}\right)^2$$
$$p = \frac{1}{10} = 0.1$$

45. If $\sqrt{3}$ = 1.732, Then what is the value of $\frac{1}{\sqrt{3}}$
(a) 0.577 (b) 2.577
(c) 1.577 (d) 0.770
RRB RPF Constable -22/01/2019 (Shift-I)

Ans : (a) Given,

$$\frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

$$= \frac{1.732}{3} = 0.57733$$

Hence, the value of $\frac{1}{\sqrt{3}} = 0.57733$.

46. If $\sqrt{45} + \sqrt{20} = 11.180$, then find the value of $\sqrt{180} + 4\sqrt{5}$

(a) 22.360 (b) 24.595
(c) 20.124 (d) 17.888

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (a) : Given,

$\sqrt{45} + \sqrt{20} = 11.180, \Rightarrow \sqrt{45} + 2\sqrt{5} = 11.180$

$\sqrt{180} + 4\sqrt{5} = 2\sqrt{45} + 4\sqrt{5}$

$2\left(\sqrt{45} + 2\sqrt{5}\right) = 2 \times 11.180 = 22.360$

47. If $3\sqrt{5} + \sqrt{125} = 17.84$ Then, how much will be $\sqrt{80} + 7\sqrt{5}$

(a) 33.3 (b) 24.53
(c) 22.0 (d) 22.3

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (b) Given,

$3\sqrt{5} + \sqrt{125} = 17.84$

$3\sqrt{5} + 5\sqrt{5} = 17.84$

$8\sqrt{5} = 17.84$

$\sqrt{5} = \frac{17.84}{8} = 2.23$

According to the question, $\sqrt{80} + 7\sqrt{5}$

$\sqrt{16 \times 5} + 7\sqrt{5} = 4\sqrt{5} + 7\sqrt{5} = 11\sqrt{5}$

Putting the value of $\sqrt{5}$

$\sqrt{80} + 7\sqrt{5} = 11 \times 2.23 = 24.53$

48. If $\sqrt{50} + \sqrt{128} = \sqrt{N}$ then what is the value of N?

(a) 26 (b) 390
(c) 338 (d) 182

RRB Group-D – 31/10/2018 (Shift-II)

Ans : (c) Given,

$\sqrt{50} + \sqrt{128} = \sqrt{N}$

$\sqrt{25 \times 2} + \sqrt{64 \times 2} = \sqrt{N}$

$5\sqrt{2} + 8\sqrt{2} = \sqrt{N}$

$13\sqrt{2} = \sqrt{N}$

Squaring in both side,

$\left(13\sqrt{2}\right)^2 = \left(\sqrt{N}\right)^2$

$169 \times 2 = N$

$338 = N$

$\boxed{N = 338}$

49. The square root of 41616 is:

(a) 196 (b) 194
(c) 204 (d) 214

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (c) Square root of 41616

	204
2	41616
+2	4
40	16
+0	00
404	1616
4	1616
	× × × ×

Hence, the square root of 41616 is 204.

50. Simplify the following expression:

$7\sqrt{48} + 7\sqrt{147}$

(a) $77\sqrt{7}$ (b) $76\sqrt{3}$
(c) $76\sqrt{7}$ (d) $77\sqrt{3}$

RRB Group-D – 04/10/2018 (Shift-II)

Ans : (d) From given expression,

$7\sqrt{48} + 7\sqrt{147}$

$= 7\sqrt{16 \times 3} + 7\sqrt{49 \times 3}$

$= 7\sqrt{4 \times 4 \times 3} + 7\sqrt{7 \times 7 \times 3}$

$= 28\sqrt{3} + 49\sqrt{3}$

$= 77\sqrt{3}$

51. If $\frac{A}{\sqrt{512}} = \frac{\sqrt{162}}{A}$, find the value of A.

(a) 144 (b) $144\sqrt{2}$
(c) 288 (d) 72

RRB Group-D – 24/10/2018 (Shift-II)

Ans. (c) Given,

$$\frac{A}{\sqrt{512}} = \frac{\sqrt{162}}{A}$$

$A \times A = \sqrt{512} \times \sqrt{162}$

$A \times A = \sqrt{8 \times 8 \times 8} \times \sqrt{2 \times 9 \times 9}$

$A^2 = \sqrt{8 \times 8 \times 16 \times 9 \times 9}$

$A = 8 \times 4 \times 9$

$A = 288$

52. If $\sqrt{0.0169}\ x = 1.3$, then x = ?

(a) 10 (b) 1
(c) 100 (d) 1000

RRB Group-D – 28/11/2018 (Shift-I)

Ans : (a) Given,

$\sqrt{0.0169}\ x = 1.3$

$x = \frac{1.3}{\sqrt{0.0169}} = \frac{1.3}{.13}$

$\boxed{x = 10}$

Hence, the required value is 10.

53. If $X^2 = 841$, then what will be the value of x?

(a) 29 (b) 41
(c) 39 (d) 31

RRB Group-D – 28/11/2018 (Shift-I)

Ans : (a) Given,

$x^2 = 841$

$x^2 = 29 \times 29$

$x = \sqrt{29 \times 29}$

$x = 29$

Hence, the value of x is 29.

54. The value of $\sqrt{214+\sqrt{107+\sqrt{196}}}$

(a) 23 (b) 15
(c) 24 (d) 18

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (b) : $\sqrt{214+\sqrt{107+\sqrt{196}}}$

$\Rightarrow \sqrt{214+\sqrt{107+14}}$

$\Rightarrow \sqrt{214+\sqrt{121}}$

$\Rightarrow \sqrt{214+11}$

$\Rightarrow \sqrt{225}$

$\Rightarrow \sqrt{15\times15}$

$= 15$

55. Solve the following:

$\sqrt{(8+2\sqrt{15})(8-2\sqrt{15})}$

(a) 1 (b) 2
(c) 3 (d) 4

RRB Group-D – 27/11/2018 (Shift-III)

Ans. (b) Given,

$\sqrt{(8+2\sqrt{15})(8-2\sqrt{15})}$

from $(a^2-b^2)=(a+b)(a-b)$

$\sqrt{(8)^2-(2\sqrt{15})^2}$

$=\sqrt{64-4\times15}$

$=\sqrt{64-60}=\sqrt{4}=2$

56. If $\sqrt{54}+\sqrt{150}=a$, What will be the value of $\sqrt{96}+\sqrt{216}$

(a) 1.20a (b) 1.50a
(c) 1.60a (d) 1.25a

RRB Group-D – 16/11/2018 (Shift-I)

Ans : (d) Given,

$\sqrt{54}+\sqrt{150}=a$

$\sqrt{2\times3\times3\times3}+\sqrt{2\times3\times5\times5}=a$

$3\sqrt{6}+5\sqrt{6}=a$

$8\sqrt{6}=a$

$\boxed{\sqrt{6}=\frac{a}{8}}$ ...(i)

According to the question,

$\sqrt{96}+\sqrt{216}=?$

$\sqrt{2\times2\times2\times2\times2\times3}+\sqrt{2\times2\times2\times3\times3\times3}$

$4\sqrt{6}+6\sqrt{6}$

$\Rightarrow 10\sqrt{6}$ ...(ii)

From equation (i) and (ii),

$\Rightarrow \frac{a}{8}\times10 \Rightarrow \frac{5a}{4}=1.25a$

57. If $\sqrt{324}=x8$, then what will the value of x?

(a) 3 (b) 2
(c) 1 (d) 4

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (c) : $\sqrt{324}=x8$

Squaring in both side,

$324=(x8)^2$

Putting x = 1,

$324=(18)^2$

$324=324$

58. Simplify –

$4\sqrt{18}+7\sqrt{32}-2\sqrt{50}$

(a) $30\sqrt{2}$ (b) $32\sqrt{3}$
(c) $36\sqrt{2}$ (d) $30\sqrt{3}$

RRB NTPC 04.04.2016 Shift : 1

Ans : (a) Given expression,

$4\sqrt{18}+7\sqrt{32}-2\sqrt{50}$

$=4\sqrt{3\times3\times2}+7\sqrt{4\times4\times2}-2\sqrt{5\times5\times2}$

$=12\sqrt{2}+28\sqrt{2}-10\sqrt{2}$

$=30\sqrt{2}$

59. If $\sqrt{x^2+y^2}=25$ and $y=2x$ then find the value of x.

(a) 5 (b) 25
(c) $\sqrt{125}$ (d) $\sqrt{5}$

RRB NTPC 28.03.2016 Shift : 2

Ans :(c) $\sqrt{x^2+y^2}=25$, $y=2x$

$\sqrt{x^2+y^2}=25$ ..... (i)

On squaring equation (i),

$x^2+y^2=625$

$x^2+(2x)^2=625$ $(\because y=2x)$

$x^2+4x^2=625$

$5x^2=625$

$x^2=125$

$x=\sqrt{125}$

60. If $x+\sqrt{x}=90$, Find x.

(a) 81 (b) 64
(c) 80 (d) 72

RRB NTPC 12.04.2016 Shift : 1

Ans : (a) Given,

$x+\sqrt{x}=90$

$\Rightarrow \sqrt{x}=90-x$

Squaring in both side,

$(\sqrt{x})^2=(90-x)^2$

$\Rightarrow x=8100+x^2-180x$

$\Rightarrow x^2 - 181x + 8100 = 0$
$\Rightarrow x^2 - 100x - 81x + 8100 = 0$
$\Rightarrow x(x-100) - 81(x-100) = 0$
$\Rightarrow (x-81)(x-100) = 0$
$\Rightarrow x = 81, 100$
From, options the value of x will be 81.

61. If $\sqrt{5} = 2.236$, then $\sqrt{5}/\sqrt{2} = ?$
(a) 1.581 (b) 1.851
(c) 2.236 (d) 1.782
RRB NTPC 29.04.2016 Shift : 2

Ans : (a) Given,
$\because \sqrt{5} = 2.236$ and as we know $\sqrt{2} = 1.414$
$$\frac{\sqrt{5}}{\sqrt{2}} = \frac{2.236}{1.414} = 1.581$$

Type - 4

62. If $\sqrt{0.0169 \times x} = 1.3$, then find the value of 'x'.
(a) 10 (b) 1000
(c) 50 (d) 100
RRB JE - 29/05/2019 (Shift-II)

Ans : (d) Given,
$$\sqrt{0.0169 \times x} = 1.3$$
$$\sqrt{\frac{169 \times x}{10000}} = \frac{13}{10}$$
Squaring in both side,
$$\frac{169 \times x}{10000} = \frac{169}{100}$$
$$x = 100$$

63. Find the square root of
$$\frac{a^2}{b^2} + \frac{b^2}{a^2} + 2$$
(a) $\frac{a}{2b} - \frac{b}{2a}$ (b) $\frac{a}{b} - \frac{b}{a}$
(c) $\frac{a-b}{2}$ (d) $\frac{a}{b} + \frac{b}{a}$
RRB JE - 02/06/2019 (Shift-II)

Ans. (d) The required square root,
$$= \sqrt{\frac{a^2}{b^2} + \frac{b^2}{a^2} + 2}$$
$$\sqrt{\left(\frac{a}{b} + \frac{b}{a}\right)^2} = \left(\frac{a}{b} + \frac{b}{a}\right)$$

64. If $\sqrt{169} = 13$, then $\frac{\left(\sqrt{.00000169}\right)}{13}$ is:
(a) 0.0013 (b) 0.001
(c) 0.0001 (d) 0.013
RRB RPF Constable -24/01/2019 (Shift-III)

Ans : (c) Given,
$\because \sqrt{169} = 13$
$$\frac{\sqrt{.00000169}}{13} = \sqrt{\frac{169}{100000000}} \times \frac{1}{13}$$
$$= \frac{13}{10000} \times \frac{1}{13} = \frac{1}{10000} = .0001$$

65. $(0.14/1.4)^2 - (0.11/1.1)^2 + (0.13/1.3)^2 = ?$
(a) 1.01 (b) 0.001
(c) 0.10 (d) 0.01
RRB RPF SI -11/01/2019 (Shift-II)

Ans : (d) From given expression,
$$\left(\frac{0.14}{1.4}\right)^2 - \left(\frac{0.11}{1.1}\right)^2 + \left(\frac{0.13}{1.3}\right)^2$$
$$= \left(\frac{14}{140}\right)^2 - \left(\frac{11}{110}\right)^2 + \left(\frac{13}{130}\right)^2$$
$$= \left(\frac{1}{10}\right)^2 - \left(\frac{1}{10}\right)^2 + \left(\frac{1}{10}\right)^2$$
$$= (0.1)^2 - (0.1)^2 + (0.1)^2$$
$$= 0.01 - 0.01 + 0.01$$
$$= 0.01$$

66. Find the value of $\sqrt{20^2 - 16^2}$
(a) 14 (b) 16
(c) 18 (d) 12
RRB RPF Constable -20/01/2019 (Shift-II)

Ans : (d) Given, $\sqrt{20^2 - 16^2}$
From- $a^2 - b^2 = (a+b)(a-b)$
$$\Rightarrow \sqrt{(20+16)(20-16)}$$
$$\Rightarrow \sqrt{36 \times 4}$$
$$\Rightarrow 6 \times 2$$
$$\Rightarrow 12$$

67. $\sqrt{\frac{?}{3136}} = \frac{1}{2}$ Find the value of '?'
(a) 56 (b) 784
(c) 1568 (d) 28
RRB Group-D – 31/10/2018 (Shift-II)

Ans : (b) Given,
$$\sqrt{\frac{?}{3136}} = \frac{1}{2}$$
Squaring in both side,
$$\left(\sqrt{\frac{?}{3136}}\right)^2 = \left(\frac{1}{2}\right)^2$$
$$\frac{?}{3136} = \frac{1}{4}$$
$$\boxed{? = 784}$$

68. What is the square root of $\frac{882}{1922}$ is:

(a) $\frac{21}{31}$ (b) $\frac{19}{31}$

(c) $\frac{22}{31}$ (d) $\frac{20}{31}$

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (a) Square root of

$$\sqrt{\frac{882}{1922}} = \sqrt{\frac{441}{961}} = \sqrt{\frac{21\times21}{31\times31}} = \frac{21}{31}$$

69. $1\frac{3}{11}$ the square of

(a) $\frac{16}{111}$ (b) $\frac{225}{112}$

(c) $\frac{196}{121}$ (d) $\frac{9}{121}$

RRB Group-D – 09/10/2018 (Shift-I)

Ans. (c) : Square root of $1\frac{3}{11}$

$$= \left(\frac{14}{11}\right)^2$$

$$= \frac{196}{121}$$

70. The value of $\left(-\sqrt{\frac{144}{576}}\right)\times\left(-\frac{16}{\sqrt{64}}\right)$

(a) 4 (b) 9

(c) 1 (d) 0

RRB Group-D – 24/10/2018 (Shift-I)

Ans : (c) Given expression,

$$\left(-\sqrt{\frac{144}{576}}\right)\times\left(-\frac{16}{\sqrt{64}}\right)$$

$$\Rightarrow \left(\frac{-12}{24}\right)\times\left(\frac{-16}{8}\right)$$

$$\Rightarrow \frac{12\times16}{24\times8} = \frac{192}{192} = 1$$

71. The value of $\sqrt{\frac{256\times289}{4^3}}$

(a) 4.25 (b) 17

(c) 8.50 (d) 34

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (d) Given expression,

$$\sqrt{\frac{256\times289}{4^3}} = \frac{16\times17}{\sqrt{4\times4\times4}} = \frac{16\times17}{2\times2\times2} = 34$$

72. $\frac{\sqrt{196}}{4.375}\times\frac{\sqrt{900}}{9.375} = ?$

(a) 8.25 (b) 8.24

(c) 10.24 (d) 9.24

Ans. (c) From given expression,

$$\Rightarrow \frac{\sqrt{196}}{4.375}\times\frac{\sqrt{900}}{9.375}$$

$$\Rightarrow \frac{14}{4.375}\times\frac{30}{9.375} = \frac{420}{41.01}$$

$$\Rightarrow 10.24$$

73. Find the value of $\frac{\sqrt{0.64}}{\sqrt{0.16}} = ?$

(a) 2 (b) 8

(c) 6 (d) 10

RRB Group-D – 02/11/2018 (Shift-II)

Ans. (a) From given expression,

$$\Rightarrow \frac{\sqrt{0.64}}{\sqrt{0.16}} = ?$$

$$= \frac{0.8}{0.4} = \frac{8}{4} = 2$$

74. The value of $\frac{\sqrt{45}\times\sqrt{20}}{\sqrt{12}\times\sqrt{3}}$

(a) 9 (b) 6

(c) 15 (d) 5

RRB Group-D – 28/09/2018 (Shift-III)

Ans : (d) Given,

$$\frac{\sqrt{45}\times\sqrt{20}}{\sqrt{12}\times\sqrt{3}} \Rightarrow \frac{3\sqrt{5}\times2\sqrt{5}}{2\sqrt{3}\times\sqrt{3}}$$

$$\Rightarrow \frac{30}{6} = \boxed{5}$$

75. Solve: $\frac{3\sqrt{121}-\sqrt{361}}{\sqrt{529}+2\sqrt{36}}$

(a) 3/5 (b) 4/7

(c) 1/4 (d) 2/5

RRB NTPC 28.03.2016 Shift : 3

Ans : (d) Given,

$$\frac{3\sqrt{121}-\sqrt{361}}{\sqrt{529}+2\sqrt{36}} = \frac{3\sqrt{11\times11}-\sqrt{19\times19}}{\sqrt{23\times23}+2\sqrt{6\times6}}$$

$$= \frac{33-19}{23+12}$$

$$= \frac{14}{35} = \frac{2}{5}$$

76. $(8)^{2/3} = ?$

(a) $\sqrt{4}$ (b) 2

(c) 4 (d) 64

RRB NTPC 11.04.2016 Shift : 1

Ans : (c) From the question,

$$8^{\frac{2}{3}} = \sqrt[3]{8^2} = \sqrt[3]{64} = \sqrt[3]{4\times4\times4} = 4$$

77. Simplify: $(25)^{\frac{3}{2}}$

(a) 625 (b) 15625

(c) 125 (d) $\sqrt{125}$

RRB NTPC 11.04.2016 Shift : 2

Ans : (c) From the given expression,

$$(25)^{\frac{3}{2}} = \left((5)^2\right)^{\frac{3}{2}} = 5^3 = 125$$

78. Simplify : $(27)^{\frac{-2}{3}}$

(a) 1/18 (b) 9
(c) 1/9 (d) 18

RRB NTPC 26.04.2016 Shift : 2

Ans : (c) $(27)^{\frac{-2}{3}} = \frac{1}{(3)^{3\times\frac{2}{3}}} = \frac{1}{(3)^2} = \frac{1}{9}$

79. $(1000)^{\frac{-1}{3}} = ?$

(a) 10 (b) 100
(c) 1/10 (d) 1/100

RRB NTPC 26.04.2016 Shift : 3

Ans : (c) $(1000)^{\frac{-1}{3}} = \frac{1}{(1000)^{\frac{1}{3}}} = \frac{1}{(10^3)^{\frac{1}{3}}} = \frac{1}{10}$

80. If $\sqrt{225} = 15$ then $\left(\sqrt{0.00000225}\right)/15 =$

(a) 0.0015 (b) 0.001
(c) 0.0001 (d) 0.015

RRB NTPC 12.04.2016 Shift : 1

Ans : (c) $\sqrt{225} = 15$

According to the question,

$$\frac{\sqrt{0.00000225}}{15}$$

$$= \frac{\sqrt{\frac{225}{100000000}}}{15}$$

$$= \frac{15}{10000\times 15} = \frac{1}{10000} = 0.0001$$

81. If $\sqrt{256} = 16$ then the value of $\frac{\sqrt{0.00000256}}{16}$ will be:

(a) 0.0016 (b) 0.001
(c) 0.0001 (d) 0.016

RRB NTPC 29.04.2016 Shift : 3

Ans. (c) Given,

$\sqrt{256} = 16$

So,

$$= \frac{\sqrt{\frac{256}{100000000}}}{16}$$

$$= \frac{16}{10000\times 16}$$

$$= \frac{1}{10000} = 0.0001$$

82. If $\sqrt{9} = 3$ then the value of $\sqrt{81}/\sqrt{3}$

(a) 3 (b) $3/\sqrt{3}$
(c) $3\sqrt{3}$ (d) 9

RRB NTPC 29.04.2016 Shift : 3

Ans. (c) Given, $\sqrt{9} = 3$

So, $\frac{\sqrt{81}}{\sqrt{3}} = \frac{9}{\sqrt{3}} = \frac{3\times\sqrt{3}\times\sqrt{3}}{\sqrt{3}} = 3\sqrt{3}$

83. If $\sqrt{144} = 12$; then $(\sqrt{.00000144})/12$:

(a) 0.0012 (b) 0.001
(c) 0.0001 (d) 0.012

RRB NTPC 30.04.2016 Shift : 1

Ans : (c) Given,

$\sqrt{144} = 12$

So,

$$\frac{\sqrt{0.00000144}}{12} = \frac{\sqrt{\frac{144}{100000000}}}{12}$$

$$= \frac{12}{10000\times 12}$$

$$= \frac{1}{10000} = 0.0001$$

Type - 5

84. What is the smallest positive integer 'x', such that $\sqrt{486x}$ is an integer?

(a) 5 (b) 6
(c) 3 (d) 2

RRB RPF Constable -17/01/2019 (Shift-I)

Ans : (b) Given,

$\sqrt{486x}$,

In order for $\sqrt{486x}$ to be an integer, $486x$ must be equal a^2 for some integer a .

$$486x = 2\times3\times3\times3\times3\times3\times x$$

$$= 2\times3^2\times3^2\times3\times x$$

$$= 3^2\times3^2\times6\times x$$

Therefore, $x = 6$ is the smallest integer that makes this the square of an integer.

85. The difference between 1/3 and 1/4 of a number is equal to its square root. Find the number.

(a) 136 (b) 144
(c) 72 (d) 120

RRB JE - 23/05/2019 (Shift-III)

Ans : (b) Let the number is x.

According to the question,

$$\frac{x}{3} - \frac{x}{4} = \sqrt{x}$$

$$\Rightarrow \frac{x}{12} = \sqrt{x}$$

Squaring in both side,

$$\frac{x^2}{144} = x$$

$$x = 144$$

86. A divisor has a reorganization number [1/(5+3√2)= ?].

(a) (5 – 2 √3)/12 (b) 5 + 2 √3/12

(c) 5 – 3 √3/12 (d) (5 – 3 √2)/7

RRB Group-D – 05/10/2018 (Shift-II)

Ans : (d) From given expression,

$$\frac{1}{5+3\sqrt{2}} = ?$$

Rationalizing the denominator,

$$= \frac{1}{5+3\sqrt{2}} \times \frac{5-3\sqrt{2}}{5-3\sqrt{2}}$$

$$= \frac{5-3\sqrt{2}}{25-18}$$

$$= \frac{(5 - 3\sqrt{2})}{7}$$

87. $\sqrt{1849} \times \sqrt{X} = 2451$ what will be the value of 'X'?

(a) 3136 (b) 3481

(c) 3364 (d) 3249

RRB Group-D – 11/10/2018 (Shift-III)

Ans : (d) Given,

$$\sqrt{1849} \times \sqrt{x} = 2451$$

$$\sqrt{43 \times 43} \times \sqrt{x} = 2451$$

$$\sqrt{x} = \frac{2451}{43}$$

$$\sqrt{x} = 57$$

Squaring in both side,

$$\left(\sqrt{x}\right)^2 = (57)^2$$

$$x = 3249$$

88. What is the value of $\sqrt{\left(3\sqrt{9} - 3\sqrt{8}\right)\left(9 + 2\sqrt{18}\right)}$

(a) 2 (b) 4

(c) 3 (d) 9

RRB Paramedical Exam – 21/07/2018 (Shift-II)

Ans : (c) From given expression,

$$\sqrt{\left(3\sqrt{9} - 3\sqrt{8}\right)\left(9 + 2\sqrt{18}\right)}$$

$$= \sqrt{\left(3 \times 3 - 6\sqrt{2}\right)\left(9 + 6\sqrt{2}\right)}$$

$$= \sqrt{\left(9 - 6\sqrt{2}\right)\left(9 + 6\sqrt{2}\right)} \quad [\because (a-b)(a+b) = a^2 - b^2]$$

$$= \sqrt{(9)^2 - \left(6\sqrt{2}\right)^2}$$

$$= \sqrt{81 - 72} = \sqrt{9} = 3$$

89. Which of the following expressions expresses the square root of $(3^{38} + 3^{39})$?

(a) $6^{38.5}$

(b) $\sqrt{2} \times 3^{19.25}$

(c) 2×3^{19}

(d) $3^{38.5}$

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (c) Given expression,

$$\sqrt{3^{38} + 3^{39}}$$

$$= \sqrt{3^{38}(1+3)}$$

$$= \sqrt{4 \times 3^{38}}$$

$$= \sqrt{2 \times 2 \times 3^{19} \times 3^{19}}$$

$$= \boxed{2 \times 3^{19}}$$

90. If the sum of square roots of two integers is $\sqrt{18+8\sqrt{5}}$. What is the sum of its squares?

(a) 164 (b) 388

(c) 624 (d) 144

RRB Group-D – 30/10/2018 (Shift-III)

Ans. (a) Let both integers are x and y.

According to the question,

$$\left(\sqrt{x} + \sqrt{y}\right) = \sqrt{18 + 8\sqrt{5}}$$

On squaring both sides,

$$\left(\sqrt{x} + \sqrt{y}\right)^2 = \left(\sqrt{18 + 8\sqrt{5}}\right)^2$$

$$x + y + 2\sqrt{xy} = 18 + 8\sqrt{5}$$

$$x + y + 2\sqrt{xy} = 10 + 8 + 2\sqrt{80}$$

$$x + y + 2\sqrt{xy} = 10 + 8 + 2\sqrt{10 \times 8}$$

Hence, on comparison,

x = 10

y = 8

So, the sum of squares,

$x^2 + y^2$

$= 10^2 + 8^2$

= 100 + 64

= 164

91. $\sqrt[3]{0.000216} = ?$

(a) 0.06 (b) 6

(c) 0.6 (d) $2\sqrt{16}$

RRB Group-D – 30/10/2018 (Shift-III)

Ans. (a) From given expression,

$\sqrt[3]{0.000216} = ?$

$\sqrt[3]{\frac{216}{1000000}}$

$\sqrt[3]{\frac{6\times6\times6}{100\times100\times100}}$

$= \frac{6}{100} = .06$

92. Which of the following expressions denotes the square root of $(3^{34} + 3^{35})$–

(a) $\sqrt{2}\times3^{17.25}$ (b) $6^{34.5}$

(c) 2×3^{17} (d) $3^{34.5}$

RRB Group-D – 31/10/2018 (Shift-I)

Ans : (c) Square root of

$(3^{34}+3^{35})$

$=\sqrt{3^{34}+3^{35}}$

$=\sqrt{3^{34}(1+3)}$

$=\sqrt{4(3^{34})} = \sqrt{2\times2\times3^{17}\times3^{17}} = 2\times3^{17}$

93. The sum of the square roots of two integers is $\sqrt{14+8\sqrt{3}}$, what is the sum of their squares?

(a) 144 (b) 388

(c) 100 (d) 162

RRB Group-D – 02/11/2018 (Shift-II)

Ans : (c) Let the integers are x and y.

According to the question,

$$\sqrt{x}+\sqrt{y} = \sqrt{14+8\sqrt{3}}$$

On squaring both sides,

$(\sqrt{x}+\sqrt{y})^2$

$=(\sqrt{14+8\sqrt{3}})^2$

$x+y+2\sqrt{xy}$

$=14+8\sqrt{3}$

On comparison,

$x+y=14$ .....(I) $2\sqrt{xy}=8\sqrt{3}$

$\sqrt{xy}=4\sqrt{3}$

$xy=48$ ..... (II)

From equation (I),

$x+y=14$

On squaring both sides,

$(x+y)^2=(14)^2$

$x^2+y^2+2xy=196$

$x^2+y^2=196-2\times48=196-96=100$

Hence, the required sum will be 100.

94. If $\sqrt{1296}=(?)^2$ then the value of (?) is:

(a) 6 (b) 18

(c) 8 (d) 12

RRB Group-D – 31/10/2018 (Shift-II)

Ans : (a) Given expression,

$\sqrt{1296}=(?)^2$

$1296=(?)^4$

$6\times6\times6\times6=?^4$

$6^4=?^4$

$\boxed{?=6}$

95. If $\sqrt{0.0361}$ x = 1.9, then x is:

(a) 1000 (b) 10

(c) 100 (d) 1

RRB Group-D – 01/12/2018 (Shift-II)

Ans : (b) Given,

$\sqrt{0.0361}\ x = 1.9$

$x=\frac{1.9}{\sqrt{0.0361}}$

$=\frac{1.9}{.19}=10$

96. A group of students decided that each member will give as much money as the number of members. If the total collection of money is Rs. 62.41, then the number of members is:

(a) 77 (b) 81

(c) 71 (d) 79

RRB Group-D – 27/11/2018 (Shift-III)

Ans : (d) Let the number of members is x.

So, received money by each member = x paise.

And total money received by members = $x\times x$ *paise*

$Rs.62.41 = Rs.\frac{x^2}{100}$ .....$(\because 1\ Rs. = 100\ paise)$

$62.41\times100 = x^2\ paise$

$x^2 = 6241\ paise$

$x=\sqrt{6241}\ paise$

$x = 79\ paise$

Hence, the number of members in group is 79.